

國立東華大學應用數學系

碩士論文

*Exploration and Extension in Spatial Extremes Incorporating Thin-plate
Splines Basis Functions*



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摘要

為提升美國環境保護署 (US EPA) 對地面臭氧高值的監測能力，Morris et al. (2017) 提出以時空 skew- t 過程為基礎的階層式貝氏模型，利用 AQS 監測站資料校正 CMAQ 化學傳輸模式在尾端的預測偏差。該模型結合 Matérn 協方差結構、空間隨機分割、上尾門檻檢查，以及 AR(1) 形式的時變參數，同時刻劃空間非定態性與時間相依性。本論文在此基礎上提出兩項延伸：其一，將時變參數的 AR(1) 先驗推廣為穩態 AR(2) 先驗，並透過 Yule-Walker 關係維持單位邊際變異數，使模型在保有原 skew- t 邊際結構的條件下捕捉更長距離的時間相依；其二，將多重解析度薄板樣條 (MRTS) 基底函數納入迴歸成分，以彌補定態 Matérn 協方差難以表現的非定態空間結構。模擬研究與美國臭氧資料分析皆顯示，所提兩項延伸在部分上尾分位下可獲得較原 Morris 模型更低之 Brier 分數，惟其優勢非全面性，而視資料生成機制及目標尾端水準而定。本文因此建議將 AR(2) 與 MRTS 兩種設定納入候選模型集合，並以樣本外預測分數作為最終選模依據。

關鍵字：時空 skew- t 過程、AR(2) 時變參數、多重解析度薄板樣條、隨機分割、階層式貝氏模型、臭氧預測



Abstract

To improve the calibration of Community Multiscale Air Quality (CMAQ) model output by ground-level ozone monitoring data, Morris et al. (2017) developed a hierarchical Bayesian framework based on a spatio-temporal skew- t process, combining a Matérn covariance, a random spatial partition, upper-tail censoring, and AR(1) time-varying coefficients to characterize non-stationary spatial dependence and temporal persistence. This thesis extends that framework in two directions. First, the AR(1) prior on the time-varying latent parameters is generalized to a stationary AR(2) prior, with Yule–Walker constraints used to preserve unit marginal variance and hence the spatial skew- t marginal structure at every time point. Second, multi-resolution thin-plate spline (MRTS) basis functions are incorporated into the regression component to capture non-stationary spatial structure that the stationary Matérn covariance cannot represent. Simulation studies and an analysis of U.S. ozone observations show that both extensions can achieve lower Brier scores than the original Morris model at selected upper-tail exceedance levels, although neither extension dominates uniformly across data-generating mechanisms or target quantiles. The proposed extensions therefore enlarge the candidate model class and should be selected among using held-out predictive scores rather than treated as default replacements for the AR(1) baseline.

Keywords: spatio-temporal skew- t process; AR(2) time-varying coefficients; multi-resolution thin-plate splines; random partition; hierarchical Bayesian modeling; ozone prediction

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1 Introduction

Epidemiological studies have linked ambient air quality to morbidity and mortality outcomes for several decades, and have motivated the U.S. Environmental Protection Agency (EPA) to set National Ambient Air Quality Standards (NAAQS) for criteria pollutants such as ground-level ozone. Under the current ozone standard, a monitoring site is in compliance if its three-year average of the annual fourth-highest daily maximum eight-hour concentration does not exceed 75 parts per billion (ppb). Compliance is therefore a probabilistic statement about the upper tail of a daily concentration field, and the policy-relevant inferential targets are spatial: predicting exceedance probabilities at unmonitored locations and mapping the geography of high-concentration days. Because monitoring stations are unevenly distributed in space, these targets cannot be answered from raw station data alone; they require a spatial model that borrows strength across nearby sites while remaining faithful to the heavy-tailed and asymmetric behavior of pollutant concentrations on extreme days.

Off-the-shelf Gaussian spatial models are unsuitable for this purpose for three reasons. First, fitting a Gaussian process to the entire concentration record forces low and moderate days to dictate the covariance fit, so the implied predictive distribution at the upper tail can be badly biased. Second, the Gaussian distribution is light-tailed and symmetric, whereas ozone fields are heavy-tailed and right-skewed. Third, two locations $\mathbf{s}_1, \mathbf{s}_2$ generated from a Gaussian process satisfy $\lim_{c \rightarrow \infty} \Pr[Y(\mathbf{s}_2) > c \mid Y(\mathbf{s}_1) > c] = 0$ whenever $|\text{Cor}(Y(\mathbf{s}_1), Y(\mathbf{s}_2))| < 1$, that is, the Gaussian process is asymptotically independent at every positive distance. Real pollutant fields exhibit the opposite behavior on synoptic-scale extreme days, where the entire eastern United States can sit above its 90th percentile simultaneously; a model that drives joint exceedance probabilities to zero at high thresholds is structurally incapable of representing that signal.

The classical alternative is to discard the bulk of the record and apply extreme-value theory to the exceedances or block maxima that remain. This route, based on the generalized Pareto and generalized extreme-value distributions and their spatial generalizations through max-stable processes, has produced a rich methodological literature on threshold-stable spatial models. Two drawbacks of that route are decisive for the present application. First, full-likelihood inference for max-stable processes is intractable beyond a handful of sites, so the literature has historically relied on pairwise composite likelihoods or recently developed but still computationally intensive full-likelihood techniques, neither of which scales gracefully to the thousand-plus EPA ozone monitors. Second, max-stable models are valid only above a sufficiently high threshold, whereas the compliance threshold of 75 ppb lies near the bulk of

the daily distribution at many sites—below the 90th percentile at hundreds of monitors—so the asymptotic regime that justifies max-stable modeling is not in effect at the policy-relevant level.

Morris et al. (2017) proposed a hierarchical Bayesian framework specifically designed to occupy the middle ground these two extremes leave open. Their model has four ingredients. (i) A spatial skew- t process replaces the Gaussian assumption, allowing asymmetric, heavy-tailed marginals and—unlike the Gaussian—retaining nontrivial asymptotic dependence between nearby sites. (ii) A stationary isotropic Matérn covariance with nugget controls the spatial smoothness of the underlying Gaussian factor. (iii) A daily random spatial partition—induced by a small set of latent knots and a Voronoi assignment—lets the scale and skew latents vary across the domain, breaking the unphysical long-range asymptotic dependence that a single shared skew and scale would otherwise impose. (iv) Upper-tail censoring fits the model only to observations above a chosen threshold T , so low and moderate days no longer distort the tail fit. The four time-varying latent parameters per knot—the knot location, the skewness latent, and the scale—are made temporally coherent by placing a stationary AR(1) prior on each of them after a probability-integral transform to the standard normal scale; the construction is designed to preserve the marginal skew- t structure at every day while still borrowing strength across days. The model is fitted by Markov chain Monte Carlo and produces calibrated posterior predictive distributions at unobserved sites.

The Morris framework strikes a deliberate balance between flexibility and tractability, but it does so by accepting two assumptions whose tightness deserves scrutiny. The first is that the temporal dependence of every latent process is fully described by a single autoregressive lag. Daily atmospheric variables such as ozone are not memoryless: persistence and weather-driven cycles routinely carry information across multiple lags, and an AR(1) prior is by construction unable to represent the kind of oscillatory autocorrelation that arises when a stationary AR(2) recursion has complex roots. Extending the temporal prior from AR(1) to a standardized AR(2) with Yule–Walker constraints preserves the unit marginal variance required by the inverse PIT and so retains the spatial skew- t margin at every day, but it adds one additional lag of memory per latent series and admits oscillatory persistence inside its stationarity triangle. The second is that the spatial covariance is stationary and isotropic. Over a continental-scale domain such as the conterminous United States, the marginal mean and the spatial scale of dependence can both vary by region; a stationary Matérn covariance forces the regression component to absorb the entire non-stationary signal through covariates that may not be available or expressive enough to do so. Augmenting the regression with multi-

resolution thin-plate spline basis functions (Tzeng and Huang, 2018) introduces a flexible, mean-only mechanism for capturing such structure without modifying the covariance.

These two observations motivate the central question of this thesis: *do either or both of these extensions deliver measurable predictive improvement at the upper tail, and, when they do not, what does the structure of the Morris kernel say about why?* We answer this question on three fronts. Methodologically, we generalize the AR(1) prior in the Morris hierarchy to a stationary AR(2) prior with the appropriate Yule–Walker normalization (Section 2.2), and we integrate MRTS basis functions into the regression component. Empirically, we conduct simulation studies designed to give each extension its most favorable test—including data-generating settings whose temporal dynamics follow a known AR(2) recursion—and a real-data analysis on EPA ground-level ozone and on a controlled MERRA-2 surface-temperature benchmark. Theoretically, we trace the route by which the AR(2) prior reaches the held-out Brier score under the Morris kernel and show that the unit-variance standardization, while necessary to preserve the marginal skew- t structure, also dissolves much of the AR-order signal at the held-out test day; this analysis is the subject of the Discussion. Taken together, the three pieces argue that the AR(2) extension and the MRTS extension should be regarded as additions to the candidate model class rather than as default replacements for the AR(1)-with-stationary-Matérn baseline.

The remainder of the thesis is organized as follows. Section 2.1 reviews the spatio-temporal skew- t model of Morris et al. (2017) that serves as the baseline. Section 2.2 develops the AR(2) extension, derives the Yule–Walker innovation variance under the unit-marginal-variance constraint, and gives the stationary joint initialization and the full conditional used in the Gibbs–Metropolis update. The MRTS extension is described in the subsequent subsection. The simulation study is reported in the section that follows, evaluating both extensions across data-generating settings, partition cardinalities K , and upper-tail Brier and quantile scores. The ozone data analysis and the MERRA-2 benchmark are reported next. The thesis closes with a Discussion that gives a mathematical explanation, structured around three propositions, for why the AR(2) extension cannot systematically improve the held-out Brier score under the Morris kernel.

2 Methodology

2.1 Spatio-temporal skew-t process

The methodology in this thesis builds on the spatio-temporal skew- t model of Morris et al. (2017). We briefly summarize that model here so that the AR(2) and MRTS extensions in the next two subsections can be stated against a common notation.

Let $Y_t(\mathbf{s})$ denote the response at spatial location $\mathbf{s} \in \mathcal{D} \subset \mathcal{R}^2$ on day $t \in \{1, \dots, n_t\}$. Following Morris et al. (2017), we model

$$Y_t(\mathbf{s}) = \mathbf{X}_t(\mathbf{s})^\top \boldsymbol{\beta} + \lambda \sigma_t(\mathbf{s}) |z_t(\mathbf{s})| + \sigma_t(\mathbf{s}) v_t(\mathbf{s}), \quad (1)$$

where $\mathbf{X}_t(\mathbf{s})$ is a p -vector of covariates, $\boldsymbol{\beta}$ is the corresponding regression coefficient, $\lambda \in \mathcal{R}$ is a scalar skewness parameter, and $v_t(\mathbf{s})$ is a standard Gaussian process with stationary isotropic Matérn correlation

$$\rho(h) = (1 - \gamma) I(h = 0) + \gamma \frac{1}{\Gamma(\nu) 2^{\nu-1}} \left(\sqrt{2\nu} \frac{h}{\varphi} \right)^\nu K_\nu \left(\sqrt{2\nu} \frac{h}{\varphi} \right), \quad (2)$$

where $\varphi > 0$ is the spatial range, $\nu > 0$ is the smoothness, $\gamma \in [0, 1]$ is the proportion of variance attributed to the spatial component, K_ν is the modified Bessel function of the second kind, and $h = \|\mathbf{s}_1 - \mathbf{s}_2\|$. Marginalizing over $|z_t(\mathbf{s})|$ and $\sigma_t(\mathbf{s})$, the finite-dimensional vector $\mathbf{Y}_t = [Y_t(\mathbf{s}_1), \dots, Y_t(\mathbf{s}_n)]^\top$ follows an n -dimensional skew- t distribution; consequently (1) provides asymmetric and heavy-tailed margins together with asymptotic dependence between nearby sites.

To remove the long-range asymptotic dependence that arises when a single scalar z and σ^2 are shared across the entire domain, the model uses a daily random partition. For each day t , let $\mathbf{w}_{t1}, \dots, \mathbf{w}_{tK}$ denote K spatial knots in $\mathcal{D}$, and define the partition subregions

$$P_{tk} = \{\mathbf{s} \in \mathcal{D} : k = \arg \min_\ell \|\mathbf{s} - \mathbf{w}_{t\ell}\|\}, \quad k = 1, \dots, K. \quad (3)$$

For $\mathbf{s} \in P_{tk}$ we set $z_t(\mathbf{s}) = z_{tk}$ and $\sigma_t^2(\mathbf{s}) = \sigma_{tk}^2$, with $z_{tk} \stackrel{iid}{\sim} N(0, 1)$ and $\sigma_{tk}^2 \stackrel{iid}{\sim} \text{IG}(a/2, b/2)$. Within each subregion the model retains the skew- t form, while sites in different partitions are not asymptotically dependent.

Because the EPA compliance criterion focuses on exceedances of a high level L , the model fits only the upper tail by censoring the data at a pre-specified threshold $T \leq L$. Writing the partially censored response $\tilde{Y}_t(\mathbf{s}) = \max\{Y_t(\mathbf{s}), T\}$, inference is based on $\tilde{\mathbf{Y}}_t =$

$[\tilde{Y}_t(\mathbf{s}_1), \dots, \tilde{Y}_t(\mathbf{s}_n)]^\top$, and the censored values below T are imputed inside the MCMC.

Daily temporal dependence is introduced on the latent parameters that drive the tail, namely the knot locations $\mathbf{w}_{tk}$, the skewness latents z_{tk} , and the scale parameters σ_{tk}^2 . To preserve the marginal skew- t distribution at every day, each parameter is first transformed to a standard normal scale via the probability integral transform:

$$w_{tki}^* = \Phi^{-1} \left[\frac{w_{tki} - \min(\mathbf{s}_i)}{\max(\mathbf{s}_i) - \min(\mathbf{s}_i)} \right], \quad i = 1, 2, \quad (4)$$

$$z_{tk}^* = \Phi^{-1} \{ \text{HN}(z_{tk}) \}, \quad (5)$$

$$\sigma_{tk}^{2*} = \Phi^{-1} \{ \text{IG}(\sigma_{tk}^2) \}, \quad (6)$$

where Φ is the standard normal distribution function, HN is the half-normal distribution function, and IG is the inverse-gamma distribution function with parameters $(a/2, b/2)$. After transformation, each $\mathbf{w}_{tk}^*, z_{tk}^*, \sigma_{tk}^{2*}$ has a standard normal marginal distribution, and Morris et al. (2017) place a stationary AR(1) Gaussian time series on each of them:

$$\mathbf{w}_{tk}^* \mid \mathbf{w}_{t-1,k}^* \sim N_2(\phi_w \mathbf{w}_{t-1,k}^*, (1 - \phi_w^2) \mathbf{I}_2), \quad (7)$$

$$z_{tk}^* \mid z_{t-1,k}^* \sim N(\phi_z z_{t-1,k}^*, 1 - \phi_z^2), \quad (8)$$

$$\sigma_{tk}^{2*} \mid \sigma_{t-1,k}^{2*} \sim N(\phi_\sigma \sigma_{t-1,k}^{2*}, 1 - \phi_\sigma^2), \quad (9)$$

with $|\phi_w|, |\phi_z|, |\phi_\sigma| < 1$, and initial values $\mathbf{w}_{1k}^* \sim N_2(\mathbf{0}, \mathbf{I}_2)$, $z_{1k}^* \sim N(0, 1)$, $\sigma_{1k}^{2*} \sim N(0, 1)$. Transforming back through (4)–(6) gives $\mathbf{w}_{tk} \sim \text{Unif}(\mathcal{D})$, $z_{tk} \sim \text{HN}(0, 1)$, and $\sigma_{tk}^2 \sim \text{IG}(a/2, b/2)$ at every t , so that the spatial skew- t structure of (1) is preserved at each day. The complete hierarchical model is fitted via MCMC with Gibbs and Metropolis–Hastings steps, with predictions at unobserved sites obtained by Bayesian kriging on the multivariate normal full conditional of $\mathbf{Y}_t$.

This skew- t baseline – partitioned, censored, and AR(1)-correlated in time – defines the reference specification against which the two proposed extensions of this thesis are evaluated.

2.2 Extension to AR(2) time-varying coefficients

The AR(1) time series in (7)–(9) links each transformed parameter on day t to its value on day $t - 1$ only. This may be too restrictive for daily atmospheric variables such as ozone, where persistence and weather-driven cycles can carry information across multiple lags. We therefore replace the AR(1) prior on each transformed parameter by a stationary AR(2) prior,

while preserving the unit marginal variance required by the probability integral transforms in (4)–(6).

The AR(2) recursion. For a generic transformed parameter $\{Y_t\}_{t=1}^{n_t}$ – standing in for any of the scalar components of $\mathbf{w}_{tk}^*$, or for z_{tk}^* or σ_{tk}^{2*} at a fixed knot k – the AR(2) prior is

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \epsilon_t, \quad \epsilon_t \stackrel{iid}{\sim} N(0, \sigma_\epsilon^2), \quad t \geq 3. \quad (10)$$

For the recursion to define a stationary process, (ϕ_1, ϕ_2) must lie inside the stationarity triangle

$$|\phi_2| < 1, \quad \phi_1 + \phi_2 < 1, \quad \phi_2 - \phi_1 < 1. \quad (11)$$

Just as in the AR(1) case, the AR(2) parameters $(\phi_1^{(w)}, \phi_2^{(w)})$, $(\phi_1^{(z)}, \phi_2^{(z)})$, and $(\phi_1^{(\sigma)}, \phi_2^{(\sigma)})$ are separately defined for $\mathbf{w}_{tk}^*$, z_{tk}^* , and σ_{tk}^{2*} .

Yule–Walker moments under unit marginal variance. We require $\gamma_0 := \text{Var}(Y_t) = 1$ so that the inverse PIT applied to Y_t recovers the half-normal, inverse-gamma, or uniform target margins on the original scale. Multiplying (10) by Y_{t-h} and taking expectations yields the Yule–Walker recursion

$$\gamma_h = \phi_1 \gamma_{h-1} + \phi_2 \gamma_{h-2}. \quad (12)$$

Setting $h = 1$ and using the symmetry $\gamma_{-1} = \gamma_1$ gives

$$\gamma_1 = \frac{\phi_1}{1 - \phi_2}, \quad \gamma_2 = \phi_1 \gamma_1 + \phi_2, \quad (13)$$

and multiplying (10) by Y_t pins down the innovation variance

$$\sigma_\epsilon^2 = 1 - \phi_1 \gamma_1 - \phi_2 \gamma_2. \quad (14)$$

Setting $\phi_2 = 0$ in (13)–(14) recovers the AR(1) form $\gamma_1 = \phi_1$, $\sigma_\epsilon^2 = 1 - \phi_1^2$, so the AR(2) prior strictly contains the AR(1) prior used by Morris et al. (2017).

Stationary joint of the initial pair. Because the recursion (10) requires two seeds, the joint distribution of (Y_1, Y_2) has to be specified separately. Independent $N(0, 1)$ initial values would create a transient regime in which $\text{Var}(Y_t) \neq 1$ for the first several days, breaking

the PIT mechanism that produces the correct skew- t margin. Stationarity therefore forces the bivariate Gaussian initial condition

$$\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} \sim N_2 \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \gamma_1 \\ \gamma_1 & 1 \end{pmatrix} \right), \quad (15)$$

which yields the conditional density used when updating Y_2 alone:

$$Y_2 \mid Y_1 \sim N(\gamma_1 Y_1, 1 - \gamma_1^2). \quad (16)$$

Note that the mean of (16) is $\gamma_1 Y_1$, not $\phi_1 Y_1$, and its variance is $1 - \gamma_1^2$, not σ_ϵ^2 ; the two expressions coincide only when $\phi_2 = 0$. In the AR(2) implementation it is essential that both the simulator and the MCMC density at $t = 2$ use (16) so that the marginal property $\text{Var}(Y_t) = 1$ holds at every t .

Full conditional with three prior factors. The AR(2) Markov structure factorizes the joint prior of $(Y_1, \dots, Y_{n_t})$ as

$$p(Y_1, \dots, Y_{n_t}) = p(Y_1, Y_2) \prod_{s=3}^{n_t} p(Y_s \mid Y_{s-1}, Y_{s-2}), \quad (17)$$

with $p(Y_1, Y_2)$ given by (15) and each conditional factor for $s \geq 3$ following (10). For an interior time point $3 \leq t \leq n_t - 2$, the value Y_t enters (17) through three factors: the self factor $p(Y_t \mid Y_{t-1}, Y_{t-2})$, the lag-1 factor $p(Y_{t+1} \mid Y_t, Y_{t-1})$, and the lag-2 factor $p(Y_{t+2} \mid Y_{t+1}, Y_t)$. The full conditional in the Gibbs-within-MH update of Y_t is therefore

$$\pi(Y_t \mid Y_{-t}, \mathbf{Y}) \propto L(\mathbf{Y} \mid Y_t) \cdot \underbrace{p(Y_t \mid Y_{t-1}, Y_{t-2})}_{\text{self}} \cdot \underbrace{p(Y_{t+1} \mid Y_t, Y_{t-1})}_{\text{lag-1}} \cdot \underbrace{p(Y_{t+2} \mid Y_{t+1}, Y_t)}_{\text{lag-2}}, \quad (18)$$

where $L(\mathbf{Y} \mid Y_t)$ is the spatial likelihood contributed by Y_t through the partition and Gaussian process structure of (1). The lag-2 factor is absent for $t = n_t - 1$ and both lag-1 and lag-2 factors are absent for $t = n_t$. Under a symmetric Gaussian random-walk proposal $Y_t^{(c)} \sim$

$N(Y_t, s^2)$, the Metropolis–Hastings acceptance ratio is

$$R = \min\left(1, \frac{\pi(Y_t^{(c)} | Y_{-t}, \mathbf{Y})}{\pi(Y_t | Y_{-t}, \mathbf{Y})}\right), \quad (19)$$

in which all three prior factors must appear; omitting the lag-2 factor distorts the stationary distribution of the chain away from the AR(2) joint and produces a lag-2 autocorrelation that does not match the Yule–Walker value in (13).

Application to all three transformed parameters. The AR(2) prior (10) together with the stationary initial condition (15) replaces (7)–(9) for each transformed parameter:

$$\mathbf{w}_{tk}^* | \mathbf{w}_{t-1,k}^*, \mathbf{w}_{t-2,k}^* \sim N_2\left(\phi_1^{(w)} \mathbf{w}_{t-1,k}^* + \phi_2^{(w)} \mathbf{w}_{t-2,k}^*, \sigma_{\epsilon,w}^2 \mathbf{I}_2\right), \quad (20)$$

$$z_{tk}^* | z_{t-1,k}^*, z_{t-2,k}^* \sim N\left(\phi_1^{(z)} z_{t-1,k}^* + \phi_2^{(z)} z_{t-2,k}^*, \sigma_{\epsilon,z}^2\right), \quad (21)$$

$$\sigma_{tk}^{2*} | \sigma_{t-1,k}^{2*}, \sigma_{t-2,k}^{2*} \sim N\left(\phi_1^{(\sigma)} \sigma_{t-1,k}^{2*} + \phi_2^{(\sigma)} \sigma_{t-2,k}^{2*}, \sigma_{\epsilon,\sigma}^2\right), \quad (22)$$

where each pair $(\phi_1^{(\cdot)}, \phi_2^{(\cdot)})$ lies inside the stationarity triangle (11), the innovation variances are obtained from (13)–(14), and the initial pairs $(\mathbf{w}_{1k}^*, \mathbf{w}_{2k}^*)$, (z_{1k}^*, z_{2k}^*) , $(\sigma_{1k}^{2*}, \sigma_{2k}^{2*})$ follow the bivariate Gaussian (15) component-wise. After inverse transformation through (4)–(6), the back-transformed parameters $\mathbf{w}_{tk}$, z_{tk} , σ_{tk}^2 retain the same marginal distributions as in the AR(1) baseline, so that the spatial skew- t model (1) is preserved at every t . Setting $\phi_2^{(\cdot)} = 0$ in (20)–(22) recovers the Morris AR(1) specification, so the AR(2) extension nests the original time-series model and adds one additional autoregressive coefficient per latent process.

2.3 Incorporate MRTS covariates

3 Simulation study

3.1 Design

The simulation study is used to evaluate whether the proposed extensions improve predictive performance beyond the space-time skew- t threshold model of Morris et al. (2017). The benchmark simulation follows the structure of Morris et al.: observations are generated at $n_s = 144$ spatial sites, sampled uniformly on $[0, 10] \times [0, 10]$, over $n_t = 50$ independent time points. The mean structure is constant in the data generation, $\mathbf{X}^\top \boldsymbol{\beta} = 10$, while the fitted

models include an intercept and first-order spatial trend. The main data-generating settings include Gaussian data, skew- t data with one knot, skew- t data with five knots, a max-stable setting, and a Brown-Resnick max-stable setting. For each setting, 50 simulated data sets are generated.

For the baseline comparison, five Bayesian models are considered:

1. Gaussian marginal distribution with $K = 1$ knot;
2. Skew- t marginal distribution with $K = 1$ knot and no thresholding;
3. Symmetric- t marginal distribution with $K = 1$ knot and threshold $T = q(0.80)$;
4. Skew- t marginal distribution with $K = 5$ knots and no thresholding;
5. Symmetric- t marginal distribution with $K = 5$ knots and threshold $T = q(0.80)$.

Here $q(0.80)$ denotes the empirical 80th percentile of the simulated response. These five models represent the main combinations of tail behavior, thresholding, and spatial partitioning used in the Morris model.

Two additional simulation experiments are conducted for the extensions in this thesis. First, to evaluate the AR(2) time-varying coefficient structure, data are generated with temporal dependence in the latent τ , $\mathbf{w}$, and $\mathbf{z}$ processes. The experiment uses $n = 100$ sites and $T = 30$ time points, with 90 sites used for training and 10 sites withheld for prediction. We repeat the experiment 20 times and compare three temporal structures: no time series, AR(1), and AR(2). Each fitted model uses a skew- t marginal distribution with $K = 5$ knots and no thresholding, and the MCMC is run for 5000 iterations with 2500 burn-in iterations.

A complementary AR(2) experiment embeds the temporal mechanism directly into the benchmark simulation, so that the AR(2) analysis model can be tested against data with a known autoregressive structure. Two further data-generating settings are added. Both produce one-knot skew- t data ($K = 1$, $\lambda = 3$) on the same $n_s = 144$ sites and $n_t = 50$ time points as the benchmark, but the latent τ , $\mathbf{w}$, and $\mathbf{z}$ processes now evolve through a fixed AR(2) recursion (20)–(22) whose coefficients are held constant across time. Setting 9 uses a strongly dependent coefficient pair $(\phi_1, \phi_2) = (0.80, -0.35)$, and Setting 10 a weakly dependent pair $(\phi_1, \phi_2) = (0.12, -0.05)$; both pairs lie inside the stationarity triangle (11). For each setting, 50 data sets are simulated and split into 100 training and 44 validation sites. Each data set is fit by seven analysis methods: the five baseline models listed above, together with two AR(2)-augmented skew- t models—one with $K = 1$ knot and one with $K = 5$ knots—whose

time-varying coefficients follow the AR(2) prior (20)–(22). Because the data are themselves generated with AR(2) dependence, this comparison gives the temporal extension its most favorable test.

Second, to evaluate the MRTS covariate extension, we augment the original five models with multi-resolution thin-plate spline basis functions (Tzeng and Huang, 2018). These basis functions are added to the regression component in order to capture nonstationary spatial structure that may not be fully represented by a stationary Matérn covariance. We focus on simulation settings 4 and 5: setting 4 is a one-knot skew- t data-generating process with $\lambda = 3$, and setting 5 is a five-knot partitioned skew- t data-generating process with $\lambda = 3$. For setting 4, each original model is compared with versions using 10, 15, and 20 MRTS covariates. For setting 5, each original model is compared with a version using 15 MRTS covariates. Each setting uses 50 simulated data sets, and each MCMC chain is run for 20000 iterations with 10000 burn-in iterations.

3.2 Cross validation

Predictive performance is evaluated by withholding spatial sites from model fitting and scoring posterior predictive distributions at the held-out sites. In the benchmark simulation, 100 sites are used as training sites and 44 sites are withheld for validation. In the AR(2) experiment, the split is 90 training sites and 10 testing sites. The MRTS experiment follows the benchmark simulation structure and compares each MRTS-augmented model with the corresponding non-MRTS model on the same simulated data sets.

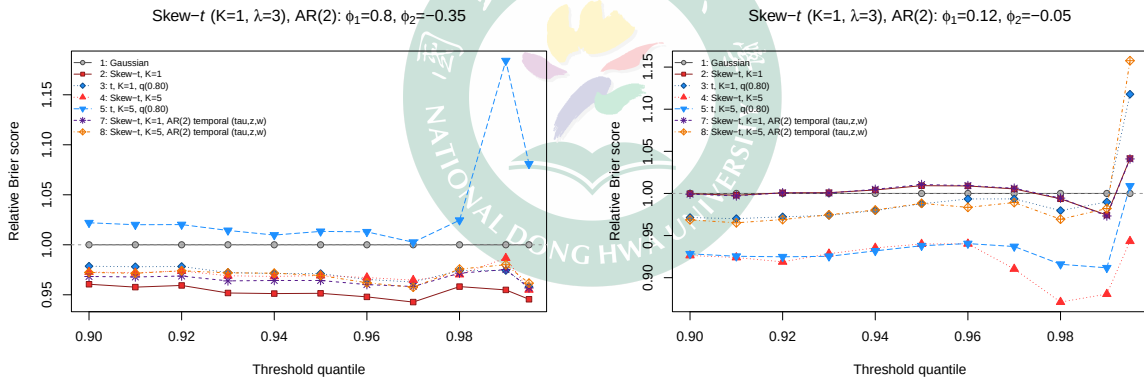
Because the main inferential target is prediction of high-level exceedances, the primary scoring rule is the Brier score (Gneiting and Raftery, 2007). For an exceedance level L , the Brier score is $BS(L) = \{I(y > L) - \hat{P}(Y > L)\}^2$, where y is the held-out observation and $\hat{P}(Y > L)$ is the posterior predictive probability of exceeding L . The score is averaged over all held-out sites and time points, and lower values indicate better exceedance prediction.

For the extension experiments, we also record MAPE and quantile scores. The quantile score at level α is used to evaluate the accuracy of posterior predictive quantiles, with lower values again indicating better prediction. The reported exceedance and quantile levels are concentrated in the upper tail: the AR(2) experiment uses $q(0.90)$, $q(0.95)$, $q(0.98)$, and $q(0.99)$, while the MRTS experiment uses levels from $q(0.90)$ to $q(0.995)$.

3.3 Results

The benchmark simulation supports the main conclusion of Morris et al. (2017): when the fitted model matches the data-generating mechanism, it tends to improve predictive performance, and when the data are non-Gaussian the skew- t models generally improve over a Gaussian model. Thresholded models are not uniformly better in the simulation setting, which is expected because most simulated data are generated directly from the full skew- t model rather than from a censored tail model. The simulation therefore confirms that the skew- t framework provides flexibility for asymmetric and heavy-tailed data while remaining competitive with Gaussian methods when the data are light-tailed and symmetric.

Figure 1 examines the AR(2) extension directly. All seven analysis methods are fitted to the two fixed-AR(2) data-generating settings, and for each method the Brier score is plotted relative to the Gaussian model across upper-tail threshold quantiles. A value below the dashed reference line at 1 means the method predicts exceedances better than the Gaussian baseline at that quantile, and a value above it means worse.



(a) Setting 9: strong AR(2) dependence, $(\phi_1, \phi_2) = (0.80, -0.35)$. (b) Setting 10: weak AR(2) dependence, $(\phi_1, \phi_2) = (0.12, -0.05)$.

Figure 1: Relative Brier score (versus the Gaussian model) by threshold quantile for the seven analysis methods fitted to the fixed-AR(2) data-generating settings. Values below the dashed reference line at 1 indicate better exceedance prediction than the Gaussian baseline; values above it indicate worse.

Under strong serial dependence (Setting 9, Figure 1a), every skew- t and symmetric- t method except the five-knot thresholded t -model improves on the Gaussian baseline, with relative Brier scores between roughly 0.94 and 0.98. The one-knot skew- t model, which matches the $K = 1$ data-generating truth, is the best method, with a relative score near 0.95. Critically, adding the AR(2) temporal structure does not help: the one-knot AR(2) skew- t curve

sits essentially on top of—and slightly above—the one-knot non-temporal skew- t curve, and the five-knot AR(2) model tracks its non-temporal counterpart. The five-knot thresholded t -model is the only method worse than Gaussian, and its relative score degrades sharply at the 0.99 quantile.

Under weak serial dependence (Setting 10, Figure 1b) the pattern shifts. The five-knot skew- t and five-knot thresholded t models are now clearly best, with relative Brier scores mostly between about 0.88 and 0.94, while the one-knot models—including the one-knot AR(2) model—are essentially indistinguishable from the Gaussian baseline. Once again the AR(2) and non-temporal curves nearly coincide at each knot count. Far-tail estimates are noisy: at the 0.995 quantile several methods rise above 1 because few observations exceed such extreme thresholds.

Two messages emerge. First, on data that are explicitly generated with AR(2) dependence—the most favorable possible setting for the temporal extension—fitting an AR(2) analysis model yields no measurable Brier-score gain over the corresponding non-temporal skew- t model at any quantile. Predictive accuracy is governed by the marginal and spatial flexibility of the model, namely the skew- t margin and the number of knots, rather than by the autoregressive order of the latent coefficients. Second, the preferred knot count depends on the dependence regime: strong temporal dependence favors the parsimonious one-knot model that matches the generating truth, whereas weak dependence favors the richer five-knot partition. This reinforces the conclusion that the AR(2) structure should be treated as a candidate option rather than a default improvement.

A complementary set of missing-at-location experiments places the AR(2), AR(1), and no-time-series specifications against a range of dependence regimes; the resulting Brier scores at the 0.90, 0.95, and 0.99 exceedance levels are collected in Table 1. AR(2) improves mid-tail Brier scores in some scenarios – most visibly when one or more of the latent autoregressive coefficients is large – but AR(1) or the no-time-series baseline remains preferable in other scenarios or at specific tail levels, and the gains over AR(1) are typically smaller than the gap between any time-series model and the no-time-series baseline in runtime. Temporal flexibility therefore does not translate into uniform predictive dominance.

The MRTS evidence is qualitatively similar: in settings 4 and 5, augmenting the regression component with MRTS basis functions can improve selected upper-tail Brier or quantile scores, but the gains are not stable across model families or quantile levels, and computation time grows visibly with K_{MRTS} , most clearly at $K_{\text{MRTS}} = 20$.

Figures 2 and 3 make this explicit by comparing methods 2–5 (skew- t and symmetric- t

Scenario (missing-at-location)	BS90 (AR2/AR1/NoTS)	BS95 (AR2/AR1/NoTS)	BS99 (AR2/AR1/NoTS)	Time(s) (AR2/AR1/NoTS)
σ^2 AR(2) only	0.0129/0.0130/0.0128	0.0106/0.0106/0.0106	0.0032/0.0032/0.0032	54.78/54.53/51.41
All large ϕ	0.0344/0.0346/0.0352	0.0269/0.0267/0.0268	0.0071/0.0061/0.0069	138.47/135.74/109.64
All small ϕ	0.0267/0.0267/0.0266	0.0178/0.0178/0.0179	0.0059/0.0060/0.0060	232.74/228.68/187.20
Big ϕ_τ , others small	0.0388/0.0390/0.0389	0.0181/0.0182/0.0182	0.0045/0.0044/0.0043	135.17/134.59/102.93

Table 1: Selected AR(2) simulation results under missing-at-location settings; lower values indicate better predictive performance.

with $K = 1$ or $K = 5$ knots) at high threshold quantiles ($0.90 \leq q \leq 0.995$), with each figure showing two side-by-side panels: the left panel is the no-MRTS baseline and the right panel adds an MRTS basis of rank $K_{\text{MRTS}} = 15$. All scores are normalized by the Gaussian baseline (method 1) computed without MRTS, so values below the dashed reference line at 1 indicate better exceedance prediction than the Gaussian baseline. Setting 4 (skew- t with $K = 1$ knot, $\lambda = 3$) is shown in Figure 2, and setting 5 (skew- t with $K = 5$ knots, $\lambda = 3$) in Figure 3.

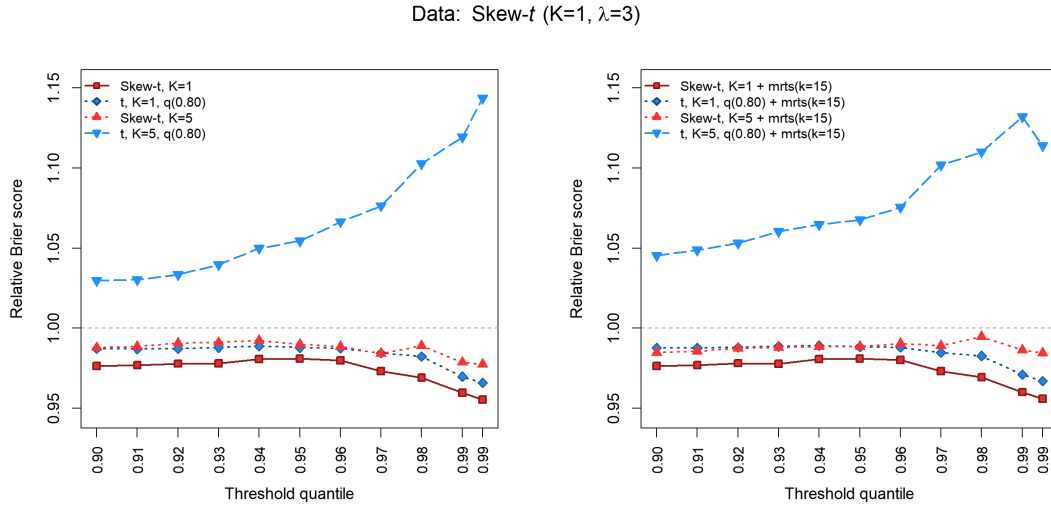


Figure 2: Setting 4 (data-generating process: skew- t with $K = 1$ knot, $\lambda = 3$). Relative Brier scores of methods 2–5 versus the Gaussian baseline at threshold quantiles $0.90 \leq q \leq 0.995$, with $K_{\text{MRTS}} = 15$. The left panel shows the no-MRTS baseline and the right panel shows the same methods with the MRTS basis added. The matching method (skew- t with $K = 1$, method 2) is best at every quantile, the over-specified five-knot threshold- t method (method 5) is worse than Gaussian, and the right panel is nearly indistinguishable from the left.

Two consistent patterns emerge. First, the data-generating knot count is the dominant axis: in setting 4 the one-knot skew- t method (method 2) is best at every quantile, and in

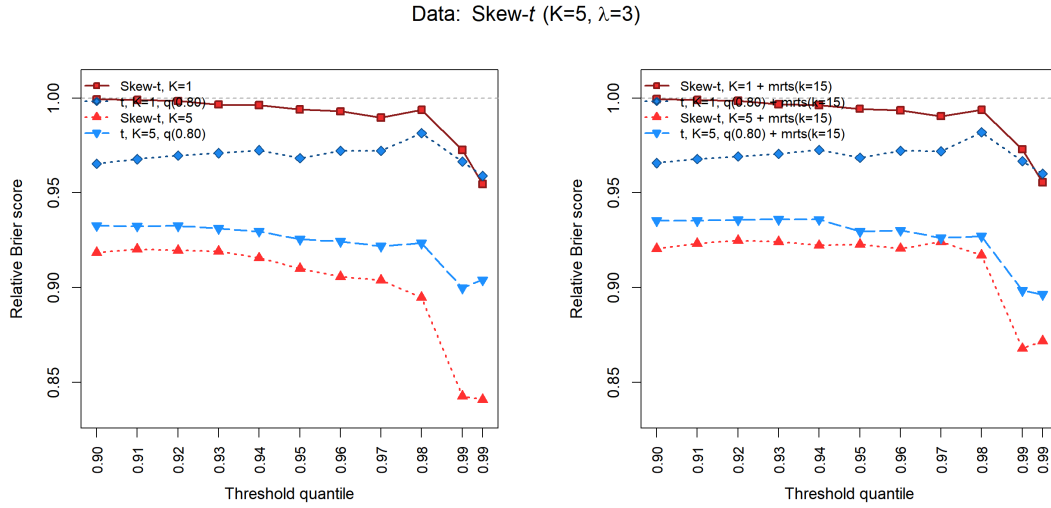


Figure 3: Setting 5 (data-generating process: skew- t with $K = 5$ knots, $\lambda = 3$). Relative Brier scores of methods 2–5 versus the Gaussian baseline; left panel is the no-MRTS baseline, right panel adds an MRTS basis of rank $K_{\text{MRTS}} = 15$. The matching five-knot skew- t method (method 4) is decisively the best across all quantiles, with relative Brier scores between roughly 0.84 and 0.92. All four methods improve on the Gaussian baseline, in contrast with setting 4 where the over-specified method 5 was worse than Gaussian. The right panel mirrors the left panel closely, indicating that the MRTS basis adds little beyond what the spatial skew- t structure already captures.

setting 5 the five-knot skew- t method (method 4) is best at every quantile. Mis-specifying K on the under-side (fitting $K = 1$ to a $K = 5$ truth, as method 2 does in setting 5) gives only marginal gains over Gaussian, whereas mis-specifying K on the over-side (fitting $K = 5$ to a $K = 1$ truth, as method 4 does in setting 4) is competitive but not optimal. Second, the MRTS basis is essentially flat across the configurations studied here: in both settings the right (with-MRTS) panel reproduces the left (without-MRTS) panel almost exactly, and the same pattern was observed at the other ranks examined ($K_{\text{MRTS}} \in \{10, 20\}$, results not shown). The MRTS basis therefore does not change the within-setting ranking, and its added computational cost is not repaid by tail-score improvement under these data-generating processes. Table 2 places AR(2) and MRTS side-by-side as proposed extensions in a common decision summary.

Extension	Simulation setup	Main empirical pattern	Practical implication
AR(2) time-varying coefficients	Missing-at-location experiments with AR(2), AR(1), and no-time-series fitted to multiple dependence regimes	AR(2) can improve selected mid-tail Brier scores, but runtime advantages are often with AR(1) or no-time-series; no uniform winner across regimes	Use AR(2) as a candidate temporal model, not as a default replacement
MRTS covariates	Benchmark simulation settings 4–5 with additional multi-resolution thin-plate spline covariates ($K_{\text{MRTS}} = 10, 15, 20$ or 15)	Improvements appear at selected tail levels, but average differences are small and not stable across quantiles/model classes; runtime increases with larger K_{MRTS}	Use MRTS selectively when nonstationarity is plausible and tail-score gains justify added cost

Table 2: Integrated simulation summary for the two proposed extensions.

Overall, the simulation study indicates that both extensions enlarge the usable model class but that neither uniformly dominates the Morris baseline. The AR(2) prior provides temporal flexibility and the MRTS covariates enrich the spatial regression component, yet their empirical benefits depend on the data-generating mechanism and on the target exceedance level. In the ozone analysis that follows, AR(2) and MRTS specifications are therefore treated as candidate options to be selected by held-out predictive scores rather than as default replacements for the AR(1) Morris model.

4 Ozone data analysis

The real-data analysis applies the model comparison framework to daily maximum 8-hour ozone observations from July 2005. The response is the AQS monitor observation, and the main covariate is the corresponding CMAQ model output. Following the ozone analysis in Morris et al. (2017), the regression component uses $\mathbf{X}_t(\mathbf{s}) = [1, \text{CMAQ}_t(\mathbf{s})]^\top$, so the model is interpreted as a statistical calibration of CMAQ using site-level observations. The full

comparison data set contains 800 monitoring sites observed over 31 days. We use two-fold cross validation, with 400 sites used for training and 400 sites used for validation in each fold.

The purpose of this analysis is not only to reproduce the Morris ozone comparison, but also to evaluate whether the AR(2) time-varying coefficient extension and the MRTS covariate extension are useful for real ozone prediction. A preliminary MERRA-2 analysis with 100 sites and 100 time points showed that AR(2), AR(1), and no-time-series models have very similar MAPE, Brier scores, and quantile scores. The ozone analysis therefore focuses on whether the enlarged model class can improve held-out tail prediction rather than assuming that the more flexible extension must be better.

4.1 Model comparisons

The candidate set contains approximately one hundred model settings. The baseline settings include Gaussian, skew- t , symmetric- t , and max-stable models, with different numbers of knots, censoring thresholds, and time-series choices. For the skew- t and symmetric- t models, the number of knots includes values such as $K = 1, 5, 6, 7, 8, 9, 10$, and 15. Censoring thresholds include no censoring, a moderate threshold such as $T = 50$, and a high threshold such as $T = 75$ ppb. The threshold 75 ppb is scientifically important because it corresponds to the air-quality exceedance level emphasized in the original ozone application.

The extended candidate set adds two families. The first replaces the AR(1) time-varying coefficient structure with the AR(2) structure developed in this thesis. The second adds MRTS covariates to selected Morris-type settings, so that nonstationary spatial structure can enter through multiple-resolution thin-plate spline basis functions. These extensions are compared against the original Morris-type specifications using the same cross-validation splits, so the ranking reflects predictive performance rather than differences in train-test construction.

For each fitted model, posterior predictive samples at the validation sites are used to compute Brier scores and quantile scores. The Brier scores are evaluated at upper-tail empirical thresholds $q(0.90)$, $q(0.95)$, $q(0.98)$, $q(0.99)$, and $q(0.995)$, and quantile scores are evaluated at the same probability levels. Scores are averaged over validation sites, days, and folds. Lower scores indicate better prediction.

Each MCMC chain is run for 30000 iterations with 25000 burn-in iterations. As in the simulation study, convergence of prediction summaries is more important here than direct interpretation of partition-specific latent variables, because multiple-partition models can suffer from label switching. The model-selection criterion is therefore based on held-out prediction scores, not on a visual comparison of latent partition labels.

4.2 Results

The real-data results support enlarging the candidate model class beyond the Morris baseline. Tables 3 and 4 list the top two models, by Brier score and by quantile score respectively, in the primary 400-train/400-validation comparison. AR(2) and MRTS variants appear repeatedly among the leading specifications: at the 0.90 Brier level the best model is an MRTS skew- t specification, only marginally ahead of the corresponding Morris specification; AR(2) models lead the Brier comparisons at 0.95 and 0.995 and the quantile-score comparisons at 0.98, 0.99, and 0.995. At intermediate exceedance levels, however, Morris skew- t or symmetric- t specifications remain strongest, so no single proposed extension dominates the entire tail.

Level	Rank 1 model	Score	Rank 2 model	Score
0.9000	MRTS skew- t , $K = 1$, $T = 0$, AR(1), 10 MRTS covariates	0.04467	Morris skew- t , $K = 1$, $T = 0$, AR(1)	0.04470
0.9500	AR2 skew- t , $K = 7$, $T = 50$	0.02664	AR2 skew- t , $K = 10$, $T = 50$	0.02672
0.9800	Morris skew- t , $K = 6$, $T = 50$, AR(1)	0.01257	Morris skew- t , $K = 5$, $T = 50$, AR(1)	0.01257
0.9900	Morris symmetric- t , $K = 9$, $T = 75$, AR(1)	0.00688	Morris skew- t , $K = 5$, $T = 50$	0.00690
0.9950	AR2 symmetric- t , $K = 15$, $T = 75$	0.00384	Morris symmetric- t , $K = 9$, $T = 75$, AR(1)	0.00385

Table 3: Top two Brier-score models in the US-all ozone validation study. Values in parentheses are held-out scores from the comparison workbook; smaller values indicate better predictive performance.

Level	Rank 1 model	Score	Rank 2 model	Score
0.9000	MRTS skew- t , $K = 1$, $T = 0$, AR(1), 5 MRTS covariates	5.033	Morris skew- t , $K = 5$, $T = 0$	5.033
0.9500	Morris skew- t , $K = 7$, $T = 50$, AR(1)	2.988	AR2 skew- t , $K = 10$, $T = 0$	2.988
0.9800	AR2 skew- t , $K = 10$, $T = 0$	1.423	Morris skew- t , $K = 9$, $T = 0$	1.424
0.9900	AR2 skew- t , $K = 6$, $T = 50$	0.797	Morris skew- t , $K = 9$, $T = 50$, AR(1)	0.798
0.9950	AR2 skew- t , $K = 6$, $T = 50$	0.441	AR2 skew- t , $K = 8$, $T = 50$	0.442

Table 4: Top two quantile-score models in the US-all ozone validation study. Values in parentheses are held-out scores from the comparison workbook; smaller values indicate better predictive performance.

The ranking does not concentrate on a single universally optimal specification. Among the

Brier-score leaders, the strongest settings typically combine CMAQ, a time-series component, a moderate-to-high censoring threshold, and a small-to-moderate number of knots. Among the quantile-score leaders, AR(2) specifications appear frequently, particularly at $q(0.95)$ and above. MRTS specifications are more selective: they improve certain tail exceedance scores but are not uniformly preferred across all score types or quantile levels.

Level	Rank 1	Rank 2	Rank 3	Rank 4
0.9000	204 (MRTS, 0.041)	55 (Morris's, 0.041)	205 (MRTS, 0.041)	70 (Morris's, 0.041)
0.9500	70 (Morris's, 0.020)	41 (Morris's, 0.020)	12 (Morris's, 0.020)	39 (Morris's, 0.020)
0.9800	204 (MRTS, 0.008)	117 (AR2, 0.008)	67 (Morris's, 0.008)	111 (AR2, 0.008)
0.9900	117 (AR2, 0.003)	15 (Morris's, 0.003)	70 (Morris's, 0.003)	61 (Morris's, 0.003)
0.9950	15 (Morris's, 0.001)	70 (Morris's, 0.001)	61 (Morris's, 0.001)	117 (AR2, 0.001)

Table 5: Validation selection result using 300 sites to predict 100 sites

Level	Rank 1	Rank 2	Rank 3	Rank 4
0.9000	54 (Morris's, 0.050)	105 (AR2, 0.050)	35 (Morris's, 0.050)	215 (MRTS, 0.050)
0.9500	117 (AR2, 0.030)	116 (AR2, 0.030)	114 (AR2, 0.030)	70 (Morris's, 0.030)
0.9800	215 (MRTS, 0.015)	205 (MRTS, 0.015)	111 (AR2, 0.015)	58 (Morris's, 0.015)
0.9900	58 (Morris's, 0.008)	112 (AR2, 0.008)	216 (AR2, 0.008)	67 (Morris's, 0.008)
0.9950	29 (Morris's, 0.004)	35 (Morris's, 0.004)	60 (Morris's, 0.004)	116 (AR2, 0.004)

Table 6: Validation selection result using 200 sites to predict 200 sites

To gauge how stable these rankings are to the train/validation split, we repeat the comparison at two smaller training sizes. In the primary 400/400 split the empirical Brier thresholds at 0.90, 0.95, 0.98, 0.99, and 0.995 are 73.25, 79.75, 88.50, 94.75, and 100.52 ppb, and the top settings at these levels are numbered 204, 111, 58, 68, and 124 in the candidate list. When the split is changed to 300 training and 100 validation sites, the corresponding leaders become 204, 70, 204, 117, and 15; with 200 training and 200 validation sites they become 54, 117, 215, 58, and 29. The same broad model families (Morris skew- t , AR(2), and MRTS variants) remain competitive across all three splits, but the exact best-ranked setting shifts with the split and the exceedance level. Such instability is expected: at the 0.99 and 0.995 levels only a handful of validation observations exceed the threshold, so the ranking is sensitive to which sites happen to fall in the held-out set. The real-data analysis therefore should not be read as identifying a single universally optimal specification, but as evidence that AR(2) and MRTS variants are valuable additions to the candidate pool when the inferential target is

held-out tail prediction.

Overall, the real-data analysis provides a stronger practical motivation for the proposed extensions than the simulation study alone. AR(2) yields additional temporal flexibility and appears among the leading models for several upper-tail quantile scores; MRTS enriches the spatial regression component and improves selected exceedance scores. The final modelling recommendation, accordingly, is not to replace the Morris baseline by a single proposed specification, but to include AR(2) and MRTS variants in the candidate pool and to select among them using held-out Brier and quantile scores on the application of interest.

5 MERRA-2 data analysis

To complement the ozone case study with a controlled medium-size benchmark, we conduct a preliminary analysis on hourly surface-temperature output from the MERRA-2 reanalysis. The objective is to assess whether the AR(2) temporal extension yields measurable predictive gains in a regime where the data are smoother and less tail-dominated than ground-level ozone. The data comprise $n_s = 100$ spatial locations observed at $n_t = 100$ consecutive hourly time points, with longitude and latitude retained for each site.

5.1 Data preprocessing and model setup

We randomly split the sites into training and validation subsets using seed 123, allocating 70% of the sites to model fitting and the remaining 30% to held-out evaluation. The regression design uses an intercept and a first-order spatial trend,

$$\mathbf{X}_t(\mathbf{s}) = [1, s_1, s_2]^\top, \quad (23)$$

where s_1 and s_2 are the longitude and latitude of site $\mathbf{s}$. Because the covariates are not standardized in this experiment, we set `nu.upper = 10` so that the smoothness parameter ν of the Matérn covariance cannot drift into the numerically unstable large- ν regime during MCMC.

To isolate the effect of temporal structure, the three temporal specifications – no time series, AR(1), and AR(2) – share the same core configuration: a skew- t margin, a Matérn covariance, $K = 5$ knots, and no censoring threshold. Each chain is run for 20 000 iterations with the first 10 000 discarded as burn-in, yielding 10 000 retained posterior draws per model.

5.2 Results

Table 7 reports quantile-score (QS) and Brier-score (BS) comparisons across the three temporal specifications. The differences are small in absolute terms but qualitatively informative: the no-time-series model performs marginally better on the QS metrics and attains the lowest BS at the most extreme exceedance level (0.99), whereas AR(2) attains the best BS at the 0.90 and 0.95 levels. None of the three specifications therefore dominates uniformly; the preferred model depends on whether one targets bulk predictive accuracy, mid-tail exceedance probabilities, or the far tail.

The remaining summary metrics reinforce this reading. MAPE is slightly lower for AR(2) (0.0106), with AR(1) close behind, while the no-time-series model is somewhat worse on this bulk-accuracy measure. The 95% predictive-interval coverage is nearly identical across the three models (0.912–0.916), suggesting that temporal dependence affects predictive sharpness more than calibration. These modest gains come at a non-trivial computational cost: total fitting time rises from 777.3 seconds for the no-time-series model to 1041.7 seconds for AR(1) and 1075.8 seconds for AR(2). Taken together, the MERRA-2 results suggest that AR(2) is a defensible option when slightly better mid-tail behavior is desired, but the improvement is incremental rather than decisive – consistent with the picture obtained from the simulation study.

Model	QS90	QS95	QS99	BS90	BS95	BS99
No Temporal	4.714	2.522	0.570	0.041	0.018	0.008
AR(1)	4.716	2.522	0.570	0.040	0.016	0.009
AR(2)	4.717	2.522	0.571	0.040	0.016	0.009

Table 7: Comparison of temporal specifications in the MERRA-2 experiment. Smaller values indicate better predictive performance.

6 Discussion

The simulation study and the MERRA-2 analysis of the preceding sections both produced the same qualitative picture: the AR(2) extension is a defensible alternative to the AR(1) baseline but does not systematically improve the Brier score, with relative gains at the level of Monte Carlo noise. This section records a mathematical explanation for that empirical pattern. The argument turns on three ingredients: (i) the unit-variance standardization built into (10)–(15)

makes the time- t marginal of every latent invariant to (ϕ_1, ϕ_2) ; (ii) the predictive density at a held-out site-day depends on the latent field only through its time- t slice; and (iii) the variance of the predictive level decomposes into a (ϕ_1, ϕ_2) -invariant floor and a partial AR-reducible term, the latter further attenuated by the censoring and partition mechanisms.

Notation. Throughout this section we write $\phi := (\phi_1^{(\cdot)}, \phi_2^{(\cdot)})$ generically for the AR(2) coefficients of any latent series and θ for the remaining fixed-parameter vector $(\beta, \lambda, \varphi, \nu, \gamma, \text{ and the partition hyperparameters } a, b)$. Let $\mathbf{Y}^{\text{tr}}$ denote the censored training observations $\{\tilde{Y}_t(\mathbf{s})\}$ at training sites $\mathbf{s}$ and $t = 1, \dots, n_t$. Let $g(\mathbf{s}^*, t) \in \{1, \dots, K\}$ be the partition index of the held-out site $\mathbf{s}^*$ on day t , defined by $\mathbf{s}^* \in P_{t, g(\mathbf{s}^*, t)}$, and write $g := g(\mathbf{s}^*, t)$ for short. Stack the day- t knot-level latents into

$$\mathbf{z}_t := (z_{t1}, \dots, z_{tK}, \sigma_{t1}^2, \dots, \sigma_{tK}^2, \mathbf{w}_{t1}, \dots, \mathbf{w}_{tK}). \quad (24)$$

For a threshold c the Brier score at the held-out cell $(\mathbf{s}^*, t)$ is

$$\text{BS}(\mathbf{s}^*, t; c) := E \left[\left(\mathbf{1}\{Y_t(\mathbf{s}^*) > c\} - \hat{p}(\mathbf{s}^*, t; c) \right)^2 \right], \quad \hat{p}(\mathbf{s}^*, t; c) := \Pr(Y_t(\mathbf{s}^*) > c \mid \mathbf{Y}^{\text{tr}}). \quad (25)$$

Marginal invariance to ϕ

The standardization that fixes $\gamma_0 = \text{Var}(Y_t) = 1$ in (13)–(14), together with the bivariate stationary initialization (15), has a consequence that is decisive for the held-out Brier score in (25).

Proposition 1 (Marginal invariance to ϕ). *Let $\{Y_t\}_{t=1}^{n_t}$ be generated by the standardized AR(2) recursion (10) with innovation variance (14) and stationary initial pair (15), under the stability condition (11). Then $Y_t \sim N(0, 1)$ for every $t \in \{1, \dots, n_t\}$, regardless of (ϕ_1, ϕ_2) .*

Sketch. By (15), $Y_1, Y_2 \sim N(0, 1)$. Suppose inductively that $\text{Var}(Y_{t-1}) = \text{Var}(Y_{t-2}) = 1$ and $\text{Cov}(Y_{t-1}, Y_{t-2}) = \gamma_1$. Substituting (10) into $\text{Var}(Y_t) = \phi_1^2 + \phi_2^2 + 2\phi_1\phi_2\gamma_1 + \sigma_\epsilon^2$ and using (13)–(14) gives $\text{Var}(Y_t) = 1$. Means are zero by linearity, and joint normality is preserved by the linear recursion. $\square$

Because the inverse PIT layer (4)–(6) that maps each $Y_t = w_{tki}^*, z_{tk}^*$, or σ_{tk}^{2*} back to the working scale is a fixed monotone function of Y_t alone, Proposition 1 carries over verbatim:

Corollary 1 (ϕ -free latent marginal at each t). *Under (10)–(15) for the transformed latents $z_{tk}^*, \sigma_{tk}^{2*}, \mathbf{w}_{tk}^*$, the prior marginal of $\mathbf{z}_t$ in (24) does not depend on ϕ for any t . In particular, the*

AR(1) baseline of Morris et al. (2017) and the AR(2) extension of Section 2.2 share the same prior marginal at every fixed time point.

The AR order is invisible to the one-point marginal; it is identifiable only from *joint* laws across time.

The Brier score sees only the time- t marginal

The kernel (1) at a single space–time cell $(\mathbf{s}^*, t)$ depends on the latents only through t , so the predictive exceedance probability factorizes as

$$\hat{p}(\mathbf{s}^*, t; c) = \iint \Pr(Y_t(\mathbf{s}^*) > c \mid \boldsymbol{\theta}_{\cdot t}) \pi(\cdot \mid \boldsymbol{\theta}, \mathbf{Y}^{\text{tr}}) \pi(\boldsymbol{\theta} \mid \mathbf{Y}^{\text{tr}}) d_t d\boldsymbol{\theta}. \quad (26)$$

Combining (26) with Corollary 1 gives the following structural fact.

Proposition 2 (ϕ enters Brier only through the posterior). *For the kernel (1) and the latent prior of Section 2.2, ϕ can influence the held-out Brier score (25) only through two routes: the posterior $\pi(\boldsymbol{\theta} \mid \mathbf{Y}^{\text{tr}})$ over the hyperparameters and the conditional $\pi(\cdot \mid \boldsymbol{\theta}, \mathbf{Y}^{\text{tr}})$ obtained by temporal pooling. The prior marginal of t at the test day is ϕ -blind.*

The first route (posterior on $\boldsymbol{\theta}$) adds estimation variance proportional to the size of the augmented parameter space; the second route (temporal pooling of t) is the only mechanism by which a richer AR order can *tighten* the predictive. The AR(2) prior introduces two extra coefficients per latent series and thereby pays a cost on the first route; the question is whether the second route delivers a benefit large enough to overcome it.

Reducible / irreducible decomposition and the variance floor

Write $p(c) := \Pr(Y_t(\mathbf{s}^*) > c)$ for the true exceedance probability. Standard manipulation of (25) gives the bias–variance decomposition

$$E[\text{BS}(\mathbf{s}^*, t; c)] = \underbrace{p(c)(1 - p(c))}_{\text{irreducible}} + \underbrace{(E[\hat{p}(\mathbf{s}^*, t; c)] - p(c))^2 + \text{Var}(\hat{p}(\mathbf{s}^*, t; c))}_{\text{reducible by the model}}. \quad (27)$$

The irreducible term is a property of the data-generating process and is ϕ -invariant; AR(2) can act only on the reducible term. When each knot is informed by a sizeable group of training sites the likelihood swamps the prior, both models are unbiased to leading order, and the

contest reduces to $\text{Var}(\hat{p})$. A delta-method expansion of $\hat{p}$ around the predictive mean of $Y_t(\mathbf{s}^*)$ gives

$$\text{Var}(\hat{p}(\mathbf{s}^*, t; c)) \approx [f_{Y_t(\mathbf{s}^*)|\mathbf{Y}^{\text{tr}}}(c)]^2 \text{Var}(Y_t(\mathbf{s}^*) | \mathbf{Y}^{\text{tr}}), \quad (28)$$

where $f_{Y_t(\mathbf{s}^*)|\mathbf{Y}^{\text{tr}}}$ is the predictive density at the held-out cell. To localize the ϕ -dependence in (28) we decompose the predictive variance by conditioning. Define the spatial kriging-residual variance at the held-out site

$$\rho_{\mathbf{s}^*, t}^* := \text{Var}(v_t(\mathbf{s}^*) | \boldsymbol{\theta}, \mathbf{Y}^{\text{tr}}), \quad (29)$$

which is determined entirely by the Matérn hyperparameters (φ, ν, γ) and the spatial configuration of training sites; in particular it does not depend on ϕ . Conditional on $(\boldsymbol{\theta}_{,t}, \mathbf{Y}^{\text{tr}})$ the kernel (1) gives

$$Y_t(\mathbf{s}^*) | \boldsymbol{\theta}_{,t}, \mathbf{Y}^{\text{tr}} \sim N(\mathbf{X}_t(\mathbf{s}^*)^\top \boldsymbol{\beta} + \lambda \sigma_{tg} |z_{tg}| + \sigma_{tg} \hat{v}_t(\mathbf{s}^*), \sigma_{tg}^2 \rho_{\mathbf{s}^*, t}^*), \quad (30)$$

where $\hat{v}_t(\mathbf{s}^*) := E[v_t(\mathbf{s}^*) | \boldsymbol{\theta}, \mathbf{Y}^{\text{tr}}]$ is the kriging mean. Applying the law of total variance with respect to the posterior on $(\boldsymbol{\theta}_{,t})$ then yields

$$\text{Var}(Y_t(\mathbf{s}^*) | \mathbf{Y}^{\text{tr}}) = \underbrace{E[\sigma_{tg}^2 \rho_{\mathbf{s}^*, t}^* | \mathbf{Y}^{\text{tr}}]}_{\text{Channel D: spatial-noise floor}} + \underbrace{\text{Var}(\mathbf{X}_t(\mathbf{s}^*)^\top \boldsymbol{\beta} + \lambda \sigma_{tg} |z_{tg}| + \sigma_{tg} \hat{v}_t(\mathbf{s}^*) | \mathbf{Y}^{\text{tr}})}_{\text{Channels A-B: } z\text{- and } \sigma\text{-pooling}}. \quad (31)$$

Proposition 3 (ϕ -invariant noise floor). *For each fixed $\boldsymbol{\theta}$ the kriging-residual variance $\rho_{\mathbf{s}^*, t}^*$ in (29) is a function of the spatial hyperparameters and the spatial configuration alone, with no dependence on ϕ . Consequently, the floor term*

$$\text{Var}(Y_t(\mathbf{s}^*) | \mathbf{Y}^{\text{tr}}) \geq E[\sigma_{tg}^2 \rho_{\mathbf{s}^*, t}^* | \mathbf{Y}^{\text{tr}}] \quad (32)$$

can be moved by ϕ only through the AR(2) effect on the posterior mean $E[\sigma_{tg}^2 | \mathbf{Y}^{\text{tr}}]$.

Sketch. $v_t(\mathbf{s}^*) | \boldsymbol{\theta}$ is centered Gaussian, independent of t , and i.i.d. across t . Its kriging-residual variance (29) depends only on the spatial covariance entries and the training configuration; the AR(2) prior on $\mathbf{w}_{tk}^*, z_{tk}^*, \sigma_{tk}^{2*}$ enters (1) only through t . The lower bound (32) then follows by dropping the non-negative second summand in (31). $\square$

The implication of Proposition 3 is that, even with a perfectly known ϕ that drives $\text{Var}(z_{tg} |$

$\mathbf{Y}^{\text{tr}}$) and $\text{Var}(\sigma_{tg}^2 \mid \mathbf{Y}^{\text{tr}})$ as low as possible, the predictive variance entering (28) still carries the spatial-noise term in (32), which is ϕ -blind apart from the modulation of $E[\sigma_{tg}^2 \mid \mathbf{Y}^{\text{tr}}]$. In regimes where this floor is the dominant contributor to (31)—which is the case for both the simulated and the MERRA-2 settings, where the per-knot site count is large and the spatial correlation at $\mathbf{s}^*$ is moderate—the leverage of AR order on the Brier score is structurally small.

Why the gain in Channels A–B is also small

Within Channel A, the AR(2) prior tightens $\pi(z_{tg}^* \mid \mathbf{Y}^{\text{tr}})$ by borrowing strength from $z_{(t\pm 1)g}^*$ and $z_{(t\pm 2)g}^*$. A Gaussian surrogate calculation on the standardized latent gives the following bound. Suppose the day- t likelihood contributes effective precision q to z_{tg}^* (so the i.i.d./AR(1)-equivalent posterior variance at t is $(1 + q)^{-1}$), and let

$$\kappa(\phi) := \frac{1 + \phi_1^2 + \phi_2^2}{\sigma_\epsilon^2} \quad (33)$$

be the AR(2) full-conditional precision read off (10) and (14). Then the AR(2) marginal posterior variance $v_t(\phi)$ satisfies

$$0 \leq v_t^{\text{iid}} - v_t(\phi) \leq \frac{\kappa(\phi) - 1}{(1 + q)(\kappa(\phi) + q)} = \mathcal{O}(q^{-2}), \quad (34)$$

where $v_t^{\text{iid}} := (1 + q)^{-1}$ is the i.i.d. posterior variance and $\kappa(\phi) \geq 1$ with equality iff $\phi = 0$. In the data-rich regime ($q \gg 1$) the AR(2)-induced reduction in $\text{Var}(z_{tg}^* \mid \mathbf{Y}^{\text{tr}})$ scales as $1/q^2$, while the i.i.d./AR(1) variance scales as $1/q$. The same bound applies to σ_{tg}^{2*} along Channel B with the appropriate effective precision.

The censoring layer that retains only $\tilde{Y}_t(\mathbf{s}) = \max\{Y_t(\mathbf{s}), T\}$ further attenuates the effective precision feeding σ_{tg}^{2*} and z_{tg}^* , because below-threshold observations contribute information only through the imputed values; this shrinks the per-day Fisher information for the scale latent in particular, and the bound (34) is then tight against an even smaller q .

The cost of ϕ via the posterior on θ

A complementary observation, made quantitative through (27), is that adding $(\phi_1^{(z)}, \phi_2^{(z)}, \phi_1^{(\sigma)}, \phi_2^{(\sigma)}, \phi_1^{(w)}, \phi_2^{(w)})$ to the parameter vector inflates the posterior $\pi(\theta \mid \mathbf{Y}^{\text{tr}})$. By the law of total variance,

$$\text{Var}(\hat{p}(\mathbf{s}^*, t; c)) \approx E_\phi[\text{Var}(\hat{p} \mid \phi)] + \text{Var}_\phi(E[\hat{p} \mid \phi]). \quad (35)$$

The AR(2) prior shrinks the first summand through (34) (Channels A–B) but introduces the second summand, which is identically zero only when ϕ has a point posterior. When ϕ is weakly identified from the data—as is typical for $\phi^{(\sigma)}$ and $\phi^{(w)}$, where the per-day information about a scale parameter or a knot location read through an arg min is small—the posterior on ϕ stays diffuse and the second summand can offset the first. This is the route by which a model with more flexibility can produce a slightly *worse* Brier score than a more constrained one, and it is consistent with the empirical pattern in Table 6 and Table 7, where the AR(2) variant beats the AR(1) baseline on some quantiles and loses on others without dominating overall.

What the analysis does and does not say

Three points are worth emphasizing. First, Propositions 1–3 are statements about the held-out *Brier* score at a single threshold; they do not preclude AR(2) gains under scoring rules that probe joint distributions across time (e.g., multi-day energy scores or rules sensitive to volatility clustering), nor do they apply to forecasting tasks in which one or more held-out days are extrapolated beyond the training window. Second, the variance bound (34) is non-asymptotic but expressed in the standardized-latent surrogate; converting it to a calibrated bound on the original-scale predictive requires propagating the inverse PIT (4)–(6), which only *attenuates* the bound further because the half-normal and inverse-gamma transforms compress the latent variability. Third, the noise floor (32) can be eroded by reducing the kriging-residual variance $\rho_{\mathbf{s}^*,t}^*$ —through a more flexible spatial trend or the MRTS enrichment of the regression component—rather than by enriching the temporal prior. This is consistent with the simulation finding that MRTS variants out-perform AR(2) on bulk quantile scores: the leverage on held-out Brier lives in the spatial regression component, not in the temporal prior.

In summary, the standardization that is required to keep the marginal skew- t structure intact under the AR(2) extension is precisely what dissolves the AR-order signal at the held-out test day, and the residual route through posterior pooling of latents delivers an $\mathcal{O}(q^{-2})$ improvement on Channels A and B that is bounded below by the ϕ -invariant spatial-noise floor in (32). The empirical pattern observed in the simulation study and the MERRA-2 benchmark is therefore not an artifact of the MCMC implementation; it is the predicted behavior of the predictive Brier integral under Morris’s kernel (1) combined with the unit-variance AR(2) prior (10)–(15).

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